

STEP 1 — Install Libraries

```
!pip install chromadb
!pip install sentence-transformers
!pip install langchain
```

```
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.11.0 in /usr/local/lib/python3.12/dist-packages)
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer-slim in /usr/local/lib/python3.12/dist-packages)
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Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages)
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Requirement already satisfied: langgraph<1.2.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from langchain-core<2.0.0,>=1.2.10 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages (from langgraph<1.2.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages (from langgraph<1.2.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: packaging>=23.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonpatch<2.0.0,>=1.33.0 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: pyyaml<7.0.0,>=5.3.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: tenacity!=8.4.0,<10.0.0,>=8.1.0 in /usr/local/lib/python3.12/dist-packages (from langsmith<1.0.0,>=0.3.45 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages (from pydantic<3.0.0,>=2.7.4 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: uuid-utils<1.0,>=0.12.0 in /usr/local/lib/python3.12/dist-packages (from typing-extensions<5.0.0,>=4.7.0 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: langgraph-checkpoint<5.0.0,>=2.1.0 in /usr/local/lib/python3.12/dist-packages (from langgraph<1.2.0,>=1.1.1 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: langgraph-prebuilt<1.1.0,>=1.0.8 in /usr/local/lib/python3.12/dist-packages (from langgraph-checkpoint<5.0.0,>=2.1.0 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: langgraph-sdk<0.4.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from langgraph-prebuilt<1.1.0,>=1.0.8 in /usr/local/lib/python3.12/dist-packages)
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Requirement already satisfied: typing-inspection>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from pydantic-core==2.41.4 in /usr/local/lib/python3.12/dist-packages)
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Requirement already satisfied: orjson>=3.11.5 in /usr/local/lib/python3.12/dist-packages (from langgraph-prebuilt<1.1.0,>=1.0.8 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: requests-toolbelt>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from langgraph-prebuilt<1.1.0,>=1.0.8 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: requests>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from langgraph-prebuilt<1.1.0,>=1.0.8 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: zstandard>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from langgraph-checkpoint<5.0.0,>=2.1.0 in /usr/local/lib/python3.12/dist-packages)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpx>=0.25.2 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0 in /usr/local/lib/python3.12/dist-packages)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.0.0 in /usr/local/lib/python3.12/dist-packages)
```

STEP 2 — Import Libraries

```
import chromadb
from sentence_transformers import SentenceTransformer
```

STEP 3 — Initialize ChromaDB

```
# Create client
client = chromadb.Client()

# Create collection (like table)
collection = client.create_collection(name="rag_collection")
```

STEP 4 — Load Embedding Model

```
model = SentenceTransformer('all-MiniLM-L6-v2')
```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF\_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>)
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or
warnings.warn(
modules.json: 100% 349/349 [00:00<00:00, 28.9kB/s]
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 11.4kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable authentication.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub.
README.md: 10.5k/? [00:00<00:00, 824kB/s]
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 5.63kB/s]
config.json: 100% 612/612 [00:00<00:00, 50.1kB/s]
model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 208MB/s]
Loading weights: 100% 103/103 [00:00<00:00, 439.46it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |
Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you
tokenizer_config.json: 100% 350/350 [00:00<00:00, 17.5kB/s]
vocab.txt: 232k/? [00:00<00:00, 8.16MB/s]
tokenizer.json: 466k/? [00:00<00:00, 21.0MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 11.9kB/s]
config.json: 100% 190/190 [00:00<00:00, 19.7kB/s]

STEP 5 — Prepare Dataset

```
documents = [  
    "Artificial Intelligence is the simulation of human intelligence.",  
    "Machine Learning is a subset of AI.",  
    "Deep Learning uses neural networks.",
```

```

    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation.",
    "AI is used in healthcare and finance.",
    "Supervised learning uses labeled data.",
    "Unsupervised learning finds hidden patterns.",
    "Reinforcement learning learns through rewards.",
    "Neural networks are inspired by the human brain.",
    "NLP helps computers understand human language.",
    "Chatbots use NLP and AI technologies.",
    "Computer vision deals with image processing.",
    "Data science involves statistics and programming.",
    "Big data requires advanced processing techniques.",
    "AI improves automation in industries.",
    "Machine learning models learn from data.",
    "Deep learning requires large datasets.",
    "RAG improves LLM accuracy using external data.",
    "Vector databases store embeddings efficiently."
]

ids = [f"doc{i+1}" for i in range(len(documents))]

```

STEP 6 — Convert Text to Embeddings

```
embeddings = model.encode(documents)
```

STEP 7 — Store in ChromaDB

```

collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)

```

STEP 8 — Perform Similarity Search

```

query = "What is Artificial Intelligence?"

# Convert query to embedding
query_embedding = model.encode([query])

# Search similar documents
results = collection.query(
    query_embeddings=query_embedding,
    n_results=3
)

print(results)

{'ids': [['doc1', 'doc2', 'doc6']], 'embeddings': None, 'documents': [['Artificial Intelligen

```

STEP 9 — Display Results

```

print("\nTop Matching Documents:\n")

for doc in results['documents'][0]:
    print("-", doc)

```

Top Matching Documents:

- Artificial Intelligence is the simulation of human intelligence.
- Machine Learning is a subset of AI.
- AI is used in healthcare and finance.

STEP 10 — Create Retrieval Function

```
def retrieve(query):  
    query_embedding = model.encode([query])  
    results = collection.query(query_embeddings=query_embedding, n_results=3)  
    return results['documents'][0]
```

STEP 11 — Test Retrieval Function

```
output = retrieve("Explain machine learning")  
  
print("\nRetrieved Documents:\n")  
for doc in output:  
    print("-", doc)
```

Retrieved Documents:

- Machine learning models learn from data.
- Machine Learning is a subset of AI.
- Supervised learning uses labeled data.

STEP 12 : Pretty Output

```
def retrieve_pretty(query):  
    print(f"\nQuery: {query}\n")  
  
    query_embedding = model.encode([query])  
    results = collection.query(query_embeddings=query_embedding, n_results=3)  
  
    print("Top Results:\n")  
    for i, doc in enumerate(results['documents'][0], 1):  
        print(f"{i}. {doc}")
```

```
retrieve_pretty("What is deep learning?")
```

Query: What is deep learning?

Top Results:

1. Deep Learning uses neural networks.
2. Deep learning requires large datasets.
3. Machine Learning is a subset of AI.

